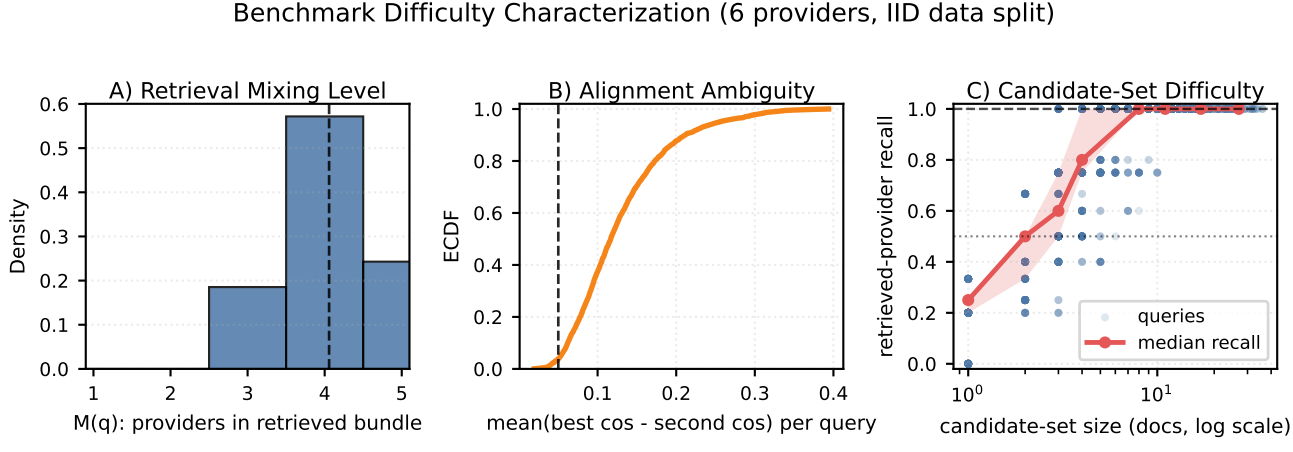
Benchmark Difficulty Characterization (6 providers, IID data split)
A) Retrieval Mixing Level
Density
M(q): providers in retrieved bundle
B) Alignment Ambiguity
ECDF
mean(best cos - second cos) per query
C) Candidate-Set Difficulty
retrieved-provider recall
candidate-set size (docs, log scale)
queries
median recall